

AWS CLOUD & DEVOPS PROJECT

Secure & Highly Available AWS Three-Tier Web Application

Architecture & Implementation Documentation

A production-inspired, highly available cloud infrastructure project spanning networking, compute, database, security, monitoring, and auditing on Amazon Web Services.

Project	Ambati — AWS Three-Tier Web Application
Architecture	3-Tier (Web / Application / Database)
Availability	2 Availability Zones
Database	Amazon RDS MySQL — Multi-AZ
Deployment	Amazon EC2 + Nginx + Apache Tomcat
Focus areas	Networking, Load Balancing, Auto Scaling, Security, Monitoring, Auditing

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1. Project Overview

The Secure & Highly Available AWS Three-Tier Web Application is a production-inspired cloud infrastructure project designed to deploy a scalable, secure, and resilient web application on Amazon Web Services (AWS).

The architecture follows a three-tier application model consisting of a Web Tier, Application Tier, and Database Tier. Each tier is logically separated using Amazon VPC networking, subnet segmentation, security groups, and load balancing.

The infrastructure is distributed across two Availability Zones to improve availability and reduce the impact of individual infrastructure failures. Public-facing traffic is routed through Amazon Route 53, AWS WAF, and an internet-facing Application Load Balancer. The Web Tier communicates with the Application Tier through an internal Application Load Balancer, while the database is hosted on Amazon RDS MySQL with Multi-AZ deployment.

Auto Scaling Groups, Amazon CloudWatch, Amazon SNS, AWS IAM, VPC Flow Logs, CloudTrail, Amazon S3, and Systems Manager Session Manager are integrated to support scalability, monitoring, security, auditing, and secure administration.

This project demonstrates practical knowledge of AWS cloud architecture, networking, compute, database services, security, load balancing, monitoring, and infrastructure operations.

Project Objective

To design and implement a secure, highly available, and scalable three-tier web application infrastructure on AWS using industry-relevant cloud services and best practices.

Key Goals

- Design a custom AWS VPC with six subnets across two Availability Zones.
- Separate Web, Application, and Database workloads.
- Deploy Nginx-based Web Tier and Tomcat-based Application Tier.
- Configure internet-facing and internal Application Load Balancers.
- Implement Auto Scaling Groups for web and application workloads.
- Configure Amazon RDS MySQL Multi-AZ for database high availability.
- Secure the infrastructure using AWS WAF, IAM, Security Groups, and private subnets.
- Implement monitoring and alerting using CloudWatch and SNS.
- Enable network traffic logging using VPC Flow Logs.
- Track AWS API activity using CloudTrail.
- Use Amazon S3 for log and static file storage.
- Access EC2 instances securely using Systems Manager Session Manager without requiring SSH keys.

2. Architecture Overview

Logical Architecture Flow

The diagram below (described as a text flow) illustrates how a user request travels through each layer of the architecture:

1. Internet — Users / Web browsers
2. Amazon Route 53 — DNS resolution
3. AWS WAF — Web application traffic protection
4. Internet-facing ALB — Public entry point (HTTPS/HTTP)
5. Web Tier (Public subnets) — EC2 + Nginx, AZ-1 and AZ-2, in a Web Auto Scaling Group
6. Internal ALB — Private load balancing
7. Application Tier (Private subnets) — EC2 + Tomcat, AZ-1 and AZ-2, in an Application Auto Scaling Group
8. Database Tier (Private DB subnets) — RDS MySQL Primary (AZ-1) and Standby (AZ-2), Multi-AZ deployment

Supporting services used throughout the architecture: NAT Gateways, CloudWatch, SNS, IAM, S3, CloudTrail, VPC Flow Logs, and Systems Manager Session Manager.

Architecture Explanation

- Users access the application through a domain name managed by Route 53.
- AWS WAF inspects web requests associated with the protected Application Load Balancer.
- The internet-facing ALB distributes traffic to healthy Nginx Web Tier instances.
- Nginx forwards application requests to the internal ALB.
- The internal ALB distributes requests to Tomcat Application Tier instances.
- Tomcat communicates with Amazon RDS MySQL using private database connectivity.
- CloudWatch monitors infrastructure health and sends notifications through SNS.
- IAM roles, security groups, and private subnets enforce access control.

Architecture note: The six subnets are distributed as two public Web subnets, two private Application subnets, and two private Database subnets. The NAT Gateways are deployed in the public subnets. The RDS Multi-AZ standby is in a separate Availability Zone, and database access is restricted to the Application Tier.

3. AWS Services and Technologies

Category	Services / Technologies	Purpose
Networking	Amazon VPC, 6 Subnets, Internet Gateway, 2 NAT Gateways	Isolated network, subnet segmentation, internet connectivity, and private outbound access.

Category	Services / Technologies	Purpose
Compute	Amazon EC2, Nginx, Apache Tomcat	Hosts the web and application workloads across multiple Availability Zones.
Database	Amazon RDS for MySQL, Multi-AZ	Managed relational database with standby deployment for availability.
Load balancing & DNS	Route 53, Internet-facing ALB, Internal ALB	Routes users to the web tier and private application requests to Tomcat.
Security & operations	AWS WAF, IAM Roles, Security Groups, CloudWatch, SNS, S3, CloudTrail, VPC Flow Logs, Session Manager	Protects, monitors, audits, and administers the environment.

4. Network Design

VPC and Subnet Plan

The following is a recommended address plan. The CIDR blocks are proposed values, not yet deployed resources.

Component	CIDR / Configuration
VPC	10.0.0.0/16
Public Web Subnet — AZ-1	10.0.1.0/24
Public Web Subnet — AZ-2	10.0.2.0/24
Private App Subnet — AZ-1	10.0.11.0/24
Private App Subnet — AZ-2	10.0.12.0/24
Private DB Subnet — AZ-1	10.0.21.0/24
Private DB Subnet — AZ-2	10.0.22.0/24

Route Table Design

Route Table	Associated Subnets	Main Routes
Public RT	Web subnet AZ-1, Web subnet AZ-2	0.0.0.0/0 → Internet Gateway
Private App RT-1	App subnet AZ-1	0.0.0.0/0 → NAT Gateway-1
Private App RT-2	App subnet AZ-2	0.0.0.0/0 → NAT Gateway-2
Private DB RT	DB subnets	Local VPC routes only

Best practice: Use separate route tables for each private application subnet and route outbound traffic through the NAT Gateway in the same Availability Zone. Database subnets should not require internet access.

Availability Zone Layout

Availability Zone	Public Web Subnet	Private App Subnet	Private DB Subnet
AZ-1	Web EC2, NAT GW-1	App EC2	RDS Primary
AZ-2	Web EC2, NAT GW-2	App EC2	RDS Standby

5. Three-Tier Application Design

Web Tier — Nginx

Technology: Amazon EC2 + Nginx

Purpose: Handles incoming web requests and serves static content or reverse-proxies application requests.

Responsibilities

- Receive traffic from the internet-facing ALB.
- Serve static content.
- Forward dynamic requests to the internal ALB.
- Participate in health checks.
- Run inside the Web Auto Scaling Group.
- Remain accessible only through the ALB.

Application Tier — Tomcat

Technology: Amazon EC2 + Apache Tomcat

Purpose: Hosts the backend Java application.

Responsibilities

- Receive requests from the internal ALB.
- Run application logic.
- Communicate with the RDS MySQL database.
- Return application responses to Nginx.
- Run in private subnets.
- Scale using an Application Auto Scaling Group.

Database Tier — RDS MySQL

Technology: Amazon RDS for MySQL, Multi-AZ

Purpose: Stores application data using a managed relational database.

Responsibilities

- Store structured application data.
- Provide managed backups and maintenance features.
- Maintain a standby database in another Availability Zone.
- Accept database connections only from the Application Tier.
- Improve availability through Multi-AZ deployment.

Note: RDS Multi-AZ is designed for high availability and failover. It is not the same as a read replica architecture for read scaling.

6. Security Architecture

Security Group Rules

Security Group	Inbound Source	Port	Purpose
ALB-SG	Internet	80 / 443	Public web traffic
Web-SG	ALB-SG	80	ALB → Nginx
App-ALB-SG	Web-SG	App listener port	Web → Internal ALB
App-SG	App-ALB-SG	8080	Internal ALB → Tomcat
DB-SG	App-SG	3306	Tomcat → MySQL

Security Principles

- No public IP addresses on application and database EC2 instances.
- No public inbound SSH access to private EC2 instances.
- Use Systems Manager Session Manager for administrative access.
- Allow only required traffic between tiers.
- Use IAM roles instead of hardcoded AWS access keys.
- Enable AWS WAF on the internet-facing ALB.
- Use HTTPS for production web traffic.
- Store secrets securely using AWS Secrets Manager or another appropriate secret management service.
- Enable encryption for RDS, EBS, and S3 as appropriate.

AWS WAF Protection

AWS WAF is configured to protect the public web entry point from unwanted or malicious web traffic. Recommended protections include:

- AWS Managed Rules.
- Common web exploit protection.
- SQL injection protection.
- Cross-site scripting protection.
- Rate-based rules where appropriate.
- Logging and monitoring of blocked requests.

7. Load Balancing and Scaling

Internet-Facing ALB

The public Application Load Balancer acts as the main entry point for users.

Flow: User → Route 53 → AWS WAF → Internet-facing ALB → Web ASG

Features

- Multi-AZ deployment.
- Listener on HTTP/HTTPS.
- Target group for Nginx EC2 instances.
- Health checks.
- Traffic distribution across healthy instances.

Internal ALB

The internal ALB is accessible only inside the VPC.

Flow: Nginx Web Tier → Internal ALB → Tomcat App ASG

Features

- Private load balancing.
- No direct public access.
- Target group for Tomcat instances.
- Health checks on the application port.
- Application tier separation.

Auto Scaling Groups

ASG	Workload	Scaling Metric
Web-ASG	EC2 + Nginx	Average CPU utilization / ALB request count

ASG	Workload	Scaling Metric
App-ASG	EC2 + Tomcat	Average CPU utilization / application metrics

Recommended Setup

- Minimum instances: 2
- Desired instances: 2
- Maximum instances: 4 or more depending on demand
- Spread instances across two Availability Zones.
- Use Launch Templates.
- Configure health checks.
- Enable scaling policies.

Note: These are proposed settings. The actual number should depend on workload, budget, and testing.

8. Monitoring, Logging, and Auditing

Amazon CloudWatch

CloudWatch monitors infrastructure and application-related metrics. Recommended monitoring:

Resource	Metrics / Monitoring
EC2	CPU utilization, status checks, network traffic
ALB	Healthy host count, request count, HTTP errors, latency
RDS	CPU, database connections, storage, free memory
NAT Gateway	Bytes processed, connection metrics
WAF	Allowed and blocked requests
CloudWatch Logs	Application and web logs

Amazon SNS

SNS sends notifications when CloudWatch alarms are triggered. Example alarms:

- Web EC2 CPU utilization above 70%.
- App EC2 CPU utilization above 70%.
- ALB unhealthy hosts.
- RDS storage or CPU threshold.

- NAT Gateway errors.
- WAF blocked request spikes.

VPC Flow Logs

VPC Flow Logs capture information about network traffic to and from network interfaces. Purpose:

- Troubleshoot connectivity.
- Investigate rejected traffic.
- Analyze network behavior.
- Support security monitoring.

Store the logs in Amazon CloudWatch Logs or Amazon S3.

AWS CloudTrail

CloudTrail records AWS API activity. Purpose:

- Track user and service actions.
- Identify resource changes.
- Support security auditing.
- Investigate unexpected modifications.

Amazon S3

Use S3 for:

- CloudTrail log storage.
- VPC Flow Logs.
- Application or web static files.
- Backup exports or other project artifacts.

9. IAM and Secure EC2 Access

IAM Roles

Create and attach appropriate IAM roles to EC2 instances. Example permissions:

- Systems Manager managed instance access.
- CloudWatch agent permissions.
- S3 access only to required buckets.
- No unnecessary administrator privileges.

Systems Manager Session Manager

Session Manager provides secure access to EC2 instances without requiring SSH keys or public IP addresses.

Benefits

- Keyless administrative access.
- No inbound SSH port required.
- Works with private EC2 instances when networking and IAM are configured.
- Supports centralized session access and auditing.

Prerequisites

- SSM Agent installed and running.
- EC2 instance IAM role with suitable Systems Manager permissions.
- Network connectivity to Systems Manager endpoints, typically through NAT or VPC endpoints.
- Correct operating system and agent configuration.

10. End-to-End Request Workflow

9. User enters the application domain name.
10. Route 53 resolves the domain to the internet-facing ALB.
11. AWS WAF inspects the incoming web request.
12. ALB forwards the request to a healthy Nginx instance.
13. Nginx serves static content or forwards dynamic traffic to the internal ALB.
14. Internal ALB selects a healthy Tomcat application instance.
15. Tomcat processes the request.
16. Tomcat communicates with RDS MySQL.
17. Database response returns to the application.
18. The response travels back through Nginx and the public ALB to the user.
19. CloudWatch monitors metrics and SNS sends alerts if configured thresholds are breached.
20. CloudTrail and VPC Flow Logs support audit and troubleshooting activities.

11. High Availability and Resilience

The project improves availability through:

- Two Availability Zones.
- Web instances distributed across AZs.
- Application instances distributed across AZs.
- Internet-facing ALB across multiple AZs.
- Internal ALB across multiple AZs.

- RDS Multi-AZ deployment.
- Two NAT Gateways for private outbound redundancy.
- Auto Scaling Groups for workload recovery and scaling.
- Health checks to route traffic away from unhealthy instances.

Failure Scenarios

Failure	Expected Behavior
One Web EC2 instance fails	ALB routes traffic to healthy instances
One App EC2 instance fails	Internal ALB routes to healthy app instances
One Availability Zone fails	Surviving AZ can continue serving if capacity and dependencies are healthy
RDS primary failure	RDS Multi-AZ failover promotes the standby
High traffic	Auto Scaling Groups can add EC2 instances
Unwanted web traffic	WAF rules can block or limit matching requests

Production consideration: True end-to-end resilience also depends on application statelessness, correct session handling, database failover testing, sufficient capacity, and the availability of all required supporting services.

12. Implementation Phases

Phase 1 — Networking

- Create VPC.
- Create six subnets across two AZs.
- Attach Internet Gateway.
- Create two NAT Gateways.
- Configure route tables.
- Enable VPC DNS support.

Phase 2 — Security

- Create security groups.
- Create IAM roles.
- Configure WAF.
- Set up Systems Manager access.
- Apply least-privilege permissions.

Phase 3 — Compute

- Launch or prepare EC2 instances.
- Install Nginx.
- Install Tomcat.
- Create Launch Templates.
- Create Web and App Auto Scaling Groups.

Phase 4 — Load Balancing

- Create public ALB.
- Create internal ALB.
- Configure listeners.
- Create target groups.
- Configure health checks.

Phase 5 — Database

- Create RDS MySQL subnet group.
- Deploy RDS MySQL Multi-AZ.
- Configure DB security group.
- Configure application database connectivity.
- Test database failover behavior.

Phase 6 — DNS and Security

- Create Route 53 record.
- Associate WAF with the public ALB.
- Configure HTTPS with ACM in production.
- Validate public and private traffic paths.

Phase 7 — Monitoring and Auditing

- Configure CloudWatch alarms.
- Configure SNS notifications.
- Enable VPC Flow Logs.
- Configure CloudTrail.
- Configure S3 log storage.
- Test alert delivery and access logging.

13. Testing and Validation

Networking Tests

- Verify VPC CIDR and subnet associations.
- Verify internet access through the public ALB.
- Verify private app outbound access through NAT Gateways.
- Verify database subnets do not have public internet routes.
- Verify route tables are correct.

Application Tests

- Open the application through the Route 53 domain.
- Verify Nginx serves the expected content.
- Verify Nginx can forward requests to Tomcat.
- Verify Tomcat can connect to RDS MySQL.
- Verify ALB target health checks.
- Verify application responses through the full three-tier path.

Security Tests

- Confirm direct access to private app and DB instances is blocked.
- Confirm only required ports are open between tiers.
- Test WAF managed rules in a safe environment.
- Verify EC2 access using Session Manager.
- Verify IAM roles provide only required permissions.

High Availability Tests

- Stop or terminate a test Web EC2 instance.
- Confirm ALB routes traffic to healthy targets.
- Test Auto Scaling behavior.
- Test RDS Multi-AZ failover according to a planned maintenance window.
- Confirm monitoring and SNS alerts.

Monitoring Tests

- Trigger a test CloudWatch alarm.
- Confirm SNS notification delivery.
- Verify CloudTrail events.
- Verify VPC Flow Logs are delivered.
- Verify S3 log storage.

14. Project Outcomes

By completing this project, the following outcomes are demonstrated:

- Designed and implemented a multi-AZ AWS VPC architecture.
- Applied three-tier application architecture.
- Deployed Nginx and Tomcat on Amazon EC2.
- Configured internet-facing and internal Application Load Balancers.
- Implemented Auto Scaling Groups.
- Configured Amazon RDS MySQL Multi-AZ.
- Applied security groups and IAM least-privilege concepts.
- Implemented AWS WAF protection.
- Configured CloudWatch monitoring and SNS alerting.
- Enabled VPC Flow Logs and CloudTrail auditing.
- Used Amazon S3 for log storage.
- Practiced secure EC2 administration with Systems Manager Session Manager.

Conclusion

The Secure & Highly Available AWS Three-Tier Web Application project demonstrates the design and implementation of a scalable, secure, and resilient cloud infrastructure using AWS services.

The architecture separates application responsibilities across Web, Application, and Database tiers while leveraging load balancing, auto scaling, private networking, managed database services, monitoring, and security controls.

This project provides practical experience relevant to Cloud Engineer, AWS Cloud Engineer, DevOps Engineer, and Junior Cloud Infrastructure Engineer roles.